

Emergency Shelter Allocation Platform: A Web-Based System to Recommend Suitable Shelters During Disasters

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Abstract—In a disaster, people need to find a safe shelter fast, but the closest shelter is not always a good choice. It may be full or closed, it may not be safe for that kind of disaster, or it may not have what the group needs. We built the Emergency Shelter Allocation Platform (ESAP), a web application that recommends shelters to a person or a family. The user enters a location, the type of disaster, a priority level, details about the group and the facilities they need. The system first removes the shelters that cannot take the group. The remaining shelters get a score out of 100 based on distance, facilities, capacity, disaster suitability and readiness, and they are ranked by this score. Administrators can add shelters, update occupancy, and keep or change the recommended shelter. We used React, Node.js, Express and PostgreSQL, with Leaflet and OpenStreetMap for the map. The scoring uses fixed rules and no machine learning, so we can always explain why a shelter was ranked where it was. In our tests on demo data for Bengaluru, a farther shelter with more space was ranked above a nearer shelter that was almost full. All shelter data in this version is demonstration data, and the project is about 50% complete.

Keywords—emergency shelter allocation, disaster management, decision support, web application, PostgreSQL, React

I. INTRODUCTION

In a flood, earthquake, fire, cyclone or landslide, many people must leave their homes and look for a safe place to stay. Shelters are set up for this, but a person in this situation often does not know which shelters are open, how many people they can still take, or what facilities they have. This information is usually not available in one place.

Going to the closest shelter seems like the obvious choice, but it can fail. The shelter may already be full or closed. It may also be unsafe for that disaster. A building that is fine in a flood can be risky in an earthquake. Some groups need things the shelter does not have, like medical help for an injured person or step-free access for someone in a wheelchair. People lose time travelling to a shelter that cannot help them, and shelters that are already crowded become more crowded.

We wanted to handle shelter selection as more than a distance search. So we built ESAP, which removes the shelters that cannot be used and ranks the rest. Each ranked shelter comes with a score breakdown. The paper first describes the existing approach and our system, and then the design, workflow, implementation, results and limitations.

II. EXISTING SYSTEM

Shelter-finding usually works as a list or a map of nearby shelters, often sorted by distance. Shelter-finding approaches generally rely on lists or map-based interfaces that present nearby emergency shelters, with distance commonly used as the primary criterion for ordering. The team studied the National Disaster Management Authority's **SACHET**

National Disaster Alert Portal and Karnataka State Disaster Management Authority's disaster-management guidelines, which provide information related to disaster response, evacuation, and identification of safe shelter locations.

Such a list does not check if a shelter can take the whole group, if it suits the disaster, or if it has the facilities the group asks for. In our problem statement we noted these gaps: it is hard to find a suitable shelter quickly, shelters can get overcrowded, travel time increases, shelter information is not in one place, capacity is not used well, and accessibility and facilities are hard to consider.

We also read ten papers on this topic. Yang et al. [1] propose CARES, which assigns routes, exits and shelters while reducing crowd conflicts, and they compare it with the nearest-shelter approach. Several papers use optimisation to allocate emergency resources under time, cost and uncertainty [2], [3], [5], [8], [9]. Li et al. [6] and Tanti et al. [7] look at where emergency facilities should be placed. Amiran et al. [4] uses machine learning to select evacuation sites, and Rajalakshmi et al. [10] discuss AI for humanitarian resource allocation. These papers mostly deal with planning models and algorithms. Our project is a working web application that handles individual requests, and it uses simple scoring instead of optimisation or machine learning.

III. PROPOSED SYSTEM

ESAP has two types of users, affected people and administrators. A user submits an emergency request with the location (GPS or a pin on the map), the disaster type, the priority, the number of people (children, senior citizens, women and disabled people are counted separately) and the facilities needed. The system gives back a ranked list of shelters with a score for each, along with a map and a driving route. The main points of the design are:

- 1) Allocation happens in two steps: hard filters first, then weighted scoring.
- 2) Each recommendation carries a score breakdown, so the user can see why a shelter was ranked where it was.
- 3) The database calculates available capacity and shelter status itself, so they cannot go out of date.
- 4) Shelters are either *registered shelters* or *potential facilities* (for example schools or community halls). A potential facility must be verified and activated by an administrator before it can be recommended.
- 5) Administrators can keep the recommended shelter or change it.

ESAP is meant to support decisions, and the final decision stays with the authorised people. All shelter records, phone numbers and occupancy figures in the system are fictional demo data.

IV. SYSTEM REQUIREMENTS

Hardware: No special hardware is needed. A normal laptop or desktop with internet and a web browser is enough, and GPS permission is optional. The software and libraries we used are listed in Table I.

TABLE I. SOFTWARE AND LIBRARIES USED

Part	Technology
Frontend	React 18, React Router 6, Vite 5
Maps	Leaflet 1.9, React-Leaflet 4, OpenStreetMap tiles
Routing	OSRM (public demo server)
Backend	Node.js 18+, Express 4
Database	PostgreSQL 14+ (PostGIS optional, not required)
Security	bcryptjs, jsonwebtoken, Helmet, CORS allowlist, express-rate-limit
Validation	Zod
Export	ExcelJS (history export to .xlsx)
Testing	Node.js test scripts, puppeteer-core
Tools	Git, GitHub, npm

V. SYSTEM DESIGN

The frontend is a React single-page app. Users get a dashboard, a new request page, a results page, request history, and a shelter list and detail page. Administrators get pages for the dashboard, shelters, requests, allocations, history and users. A Leaflet component shows the map. The browser gets map tiles from OpenStreetMap and routes from OSRM directly (Fig. 1).

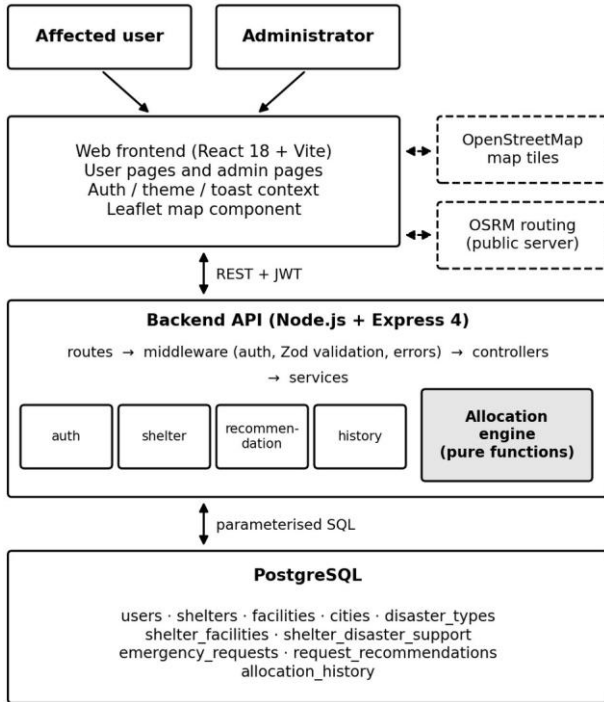


Fig. 1. Overall system architecture

The backend is an Express API. A request goes through the routes, then the middleware (authentication, role check, Zod validation, error handling), the controllers and the services. The services handle authentication, shelters, recommendations, history and capacity estimation, and there is the allocation engine. We wrote the engine as pure functions that do not touch

the database. This lets us test it on its own, and both the "save request" and "preview" endpoints use it.

The PostgreSQL database has ten tables: users, disaster_types, facilities, cities, shelters, shelter_facilities, shelter_disaster_support, emergency_requests, request_recommendations and allocation_history. Two rules are handled by the database itself. available_capacity is a generated column (total capacity minus occupancy). A trigger sets the status to *full*, *limited* (less than 20% free) or *available*, and it does not overwrite a *closed* status set by an administrator.

VI. METHODOLOGY / WORKFLOW

Fig. 2 shows the workflow. The user logs in and submits a request. The server checks the input, then loads all active shelters along with their facilities and their rating for the requested disaster. If a shelter has no rating for that disaster, we treat it as 0.45 suitable. After this, the system calculates the straight-line (haversine) distance to each shelter.

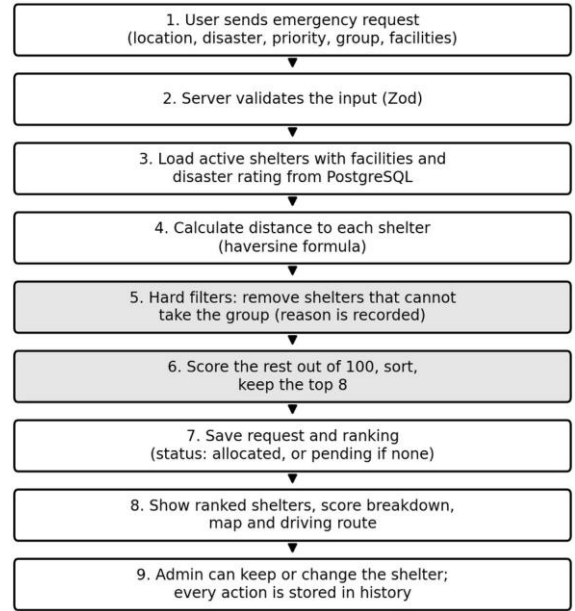


Fig. 2. Request-to-recommendation workflow

A. Hard filters

A shelter is removed and never ranked if any of these is true: it is inactive, it is not yet activated, it is closed or full, it has no free space, it is too small for the whole group, its rating for the disaster is below 0.30, it is farther than the maximum distance for that priority, or it has no medical support when a *critical* request asks for medical help. Each removed shelter is returned with a reason code.

B. Scoring

Each remaining shelter gets a score from 0 to 100:

$$S = 100 \times (w_d D + w_f F + w_c C + w_g G + w_r R) \quad (1)$$

Here D, F, C, G and R are the distance, facility, capacity, disaster-suitability and readiness scores (each between 0 and 1), and the w values are their weights (Table II).

TABLE II. SCORING FACTORS

Factor	Weight	What it checks
Distance	30%	Score drops as distance grows
Facilities	25%	Requested facilities available; medical, food, water and toilets count double
Capacity	20%	Space left after the group is admitted
Disaster suitability	20%	Rating (0–1) of the shelter for the chosen disaster
Readiness	5%	Shelter status, plus small bonuses for child, women and accessibility support when the group needs them

The priority level changes the weights, as shown in Table III, and the weights are then scaled so that they add up to 1. Results are sorted by score, then by distance, then by id, so the same input always gives the same ranking. We keep the top 8. The request and its ranking are saved together. The request becomes *allocated*, or stays *pending* if no shelter qualifies.

TABLE III. PRIORITY SETTINGS

Priority	Distance ×	Facility ×	Reach (km)	Max (km)
Critical	1.60	0.70	4	25
High	1.30	0.85	6	35
Medium	1.00	1.00	9	50
Low	0.80	1.15	12	60

VII. IMPLEMENTATION

Backend. Passwords are hashed with bcrypt and sessions use JWT. On every request the user is read again from the database, so a deactivated account loses access right away. Login and registration are limited to 20 requests per 15 minutes, and the whole API to 300 per minute. Every write endpoint is checked with Zod. If an ADMIN_PASSKEY is set, admin login needs it, and a user can register as admin only with the correct passkey.

Shelters and facilities. Administrators can create, edit and deactivate shelters and edit occupancy directly in the table. A shelter can be deleted only if no request refers to it. There are endpoints to mark a potential facility as under verification, verify it, and activate it. If a facility has no official capacity, its capacity can be estimated from the floor area at 3.5 m² per person, which is the Sphere minimum standard [11]. Such a capacity is stored and labelled as *estimated*.

Allocation review. An administrator can keep the shelter picked by the engine or change it to another shelter from that request's ranking. Every action is stored in allocation_history, and this history can be exported to Excel.

Frontend. On the new-request page the location comes from GPS or a pin on the map, so coordinates are never typed by hand. The results page shows ranked cards with a "Why this score?" panel, and a map. If OSRM cannot return a route, the page shows a warning and a link to OpenStreetMap directions. It never draws a straight line and calls it a route. The interface has light and dark themes.

Data. The seed data has 14 fictional registered shelters in Bengaluru and 4 fictional potential facilities (two potential, one under verification and one registered). It also has 10 facility types and 6 disaster types. The source code is at github.com/HarshilxAI/Emergency-Shelter-Allocation-Platform.

VIII. RESULTS

We ran requests through the preview endpoint using the demo data. For the first test, the location was a point in Indiranagar, Bengaluru. The group had 6 people (2 children, 1 senior, 2 women) who needed food, water and toilets, in a flood with medium priority. Table IV shows the top five results.

TABLE IV. TOP 5 RESULTS FOR THE SAMPLE FLOOD REQUEST

Rank	Shelter	Dist. (km)	Free spaces	Score
1	Koramangala Community Relief Centre	5.13	330	87.4
2	Indiranagar Municipal School Shelter	0.73	35 (limited)	82.9
3	Rajajinagar Sports Complex Shelter	9.68	590	82.5
4	Jayanagar Civic Assembly Hall	8.61	420	82.3
5	KR Puram Railway Colony Shelter	5.14	230	81.7

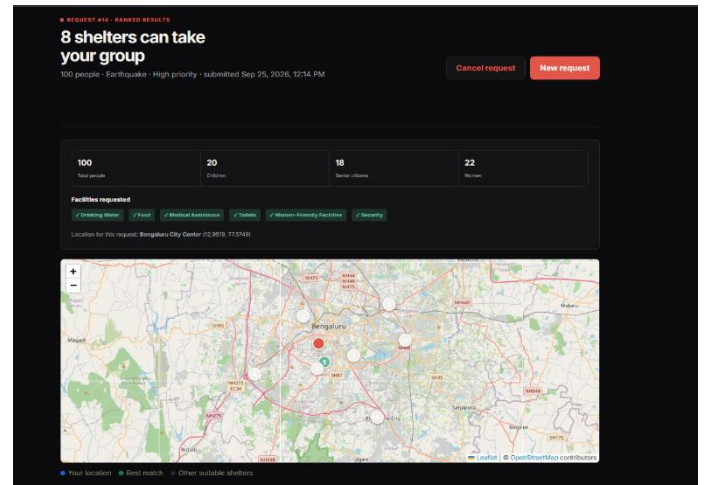
The nearest shelter, Indiranagar at 0.73 km, came second. Its distance score was 92, but its capacity score was 52.6 and its readiness score was 55 because it is almost full. Koramangala scored only 61.4 on distance, but it scored 100 on capacity, facilities and readiness. Out of the 18 shelters, 12 passed the filters and 6 were removed (1 full, 1 closed and 4 potential facilities that are not yet activated).

TABLE V. EFFECT OF CHANGING THE INPUTS

Case	Input	Top shelter (score)	Kept	Cut
A	Flood, medium, 6 people	Koramangala (87.4)	12	6
B	Flood, critical, needs medical	Koramangala (73.3)	5	13
C	Earthquake, medium	Koramangala (84.4)	11	7
D	Flood, low priority	Koramangala (91.1)	12	6
E	Flood, high, 400 people	Rajajinagar (70.5)	2	16

In case B, 7 shelters were removed because they have no medical support. In case C, one shelter was removed because it is unsuitable for earthquakes. In case D, Indiranagar moved from rank 2 to rank 8. In case E, 10 shelters were removed because they could not fit 400 people.

Testing. The 36 unit tests of the allocation engine pass. They cover the distance calculation, the hard filters, the ranking behaviour, the scoring rules and some edge cases.



Ranked recommendations

Best match first

1

Jayanagar Civic Assembly Hall

9th Block, Jayanagar East

AvailableBest matchWheelchair accessible

3.1 km

420

180 / 600

85%

Drinking WaterFoodMedical AssistanceToiletsWomen-Friendly FacilitiesSecurityElectricity

Accessibility Support

View detailsGet routeWhy this score?Call 99999-10103

2

Rajajinagar Sports Complex Shelter

Dr. Rajender Road, Rajajinagar

AvailableWheelchair accessible

5.0 km

590

210 / 600

85%

Drinking WaterFoodMedical AssistanceToiletsWomen-Friendly FacilitiesSecurityElectricity

Child-Friendly FacilitiesAccessibility SupportPet Friendly

View detailsGet routeWhy this score?Call 99999-10106

3

Koramangala Community Relief Centre

4th Block, 80 Feet Road, Koramangala

AvailableWheelchair accessible

5.9 km

330

120 / 600

80%

View detailsGet routeWhy this score?Call 99999-10103

100

20

18

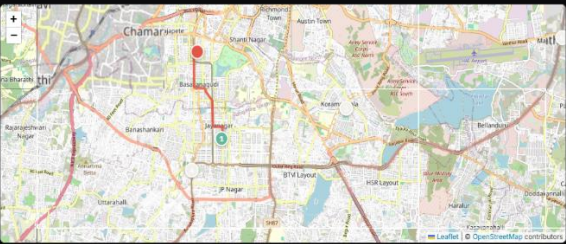
22

Total peopleChildrenSenior citizensWomen

Facilities requested

Drinking WaterFoodMedical AssistanceToiletsWomen-Friendly FacilitiesSecurity

Location for this request: Bengaluru City Center (12.919, 77.5749)



Road route found

4.0 km by road, roughly 5 minutes driving. Open turn-by-turn directions

Ranked recommendations

Best match first

1

Jayanagar Civic Assembly Hall

9th Block, Jayanagar East

AvailableBest matchWheelchair accessible

3.1 km

420

180 / 600

85%

Drinking WaterFoodMedical AssistanceToiletsWomen-Friendly FacilitiesSecurityElectricity

Accessibility Support

View detailsGet routeHide scoringCall 99999-10103

Capacity

Disaster fit

Distance

Readiness

Facilities

The overall score is the weighted average of these five factors. Weights shift with the emergency priority you selected.

Allocation Control

Review current shelter assignments and confirm or change allocations.

View pending request

9

7

22%

Active allocationsReserve capacity assignedAwaiting reviewRequests requiring system checkConfirmed allocations7 verified by admin

Active allocations

Awaiting review

Confirmed

REQUEST	GROUP	SHELTER	PRIORITY	STATUS	ALLOCATED	ACTION	
#14	Earthquake	100 people	Jayanagar Civic Assembly Hall	High	Awaiting review	2 min ago	Review
#13	Earthquake	100 people	Rajajinagar Sports Complex Shelter	High	Awaiting review	7 min ago	Review
#8	Earthquake	30 people	Jayanagar Civic Assembly Hall	High	Awaiting review	1 d ago	Review
#5	Earthquake	15 people	Rajajinagar Sports Complex Shelter	High	Awaiting review	1 d ago	Review
#2	Flood	4 people	Jayanagar Civic Assembly Hall	High	Confirmed	1 d ago	Review
#1	Flood	4 people	Jayanagar Civic Assembly Hall	High	Confirmed	1 d ago	Review
#10	Flood	10 people	Rajajinagar Sports Complex Shelter	Medium	Awaiting review	22 min ago	Review
#7	Earthquake	10 people	Rajajinagar Sports Complex Shelter	Medium	Awaiting review	1 d ago	Review

Emergency Intake

Requests

Review incoming emergency requests, user details, and allocation status.

View pending request

Search request ID, disaster, location, or user

All statuses

All priorities

REQUEST	DISASTER	GROUP	PRIORITY	STATUS	SUBMITTED	
#12	Earthquake	90 people	High	Pending	13 min ago	Open
#4	Fire	2 people	High	Pending	1 d ago	Open
#14	Earthquake	100 people	High	Allocated	7 min ago	Open
#13	Earthquake	100 people	High	Allocated	12 min ago	Open
#11	Flood	1 people	High	Cancelled	18 min ago	Open
#9	Flood	1 people	High	Cancelled	22 min ago	Open
#8	Cyclone	30 people	High	Allocated	1 d ago	Open
#5	Earthquake	15 people	High	Allocated	1 d ago	Open

Audit Trail

History

Review previous requests, allocation decisions, and administrative actions.

Export history (xlsx)

Search request, shelter, or action

All actions

From

To

DATE / TIME	REQUEST	ACTION	BY	DETAILS
Sep 25, 2025, 12:14 PM	#14	Allocated - Jayanagar Civic Assembly Hall	Demo User	Details
Sep 25, 2025, 12:09 PM	#13	Allocated - Rajajinagar Sports Complex Shelter	Demo User	Details
Sep 24, 2025, 11:45 PM	#12	Closed - Rajajinagar Sports Complex Shelter	Dharmesh Gurjar	Details
Sep 24, 2025, 11:39 PM	#12	Allocated - Rajajinagar Sports Complex Shelter	Hareesh Gurjar	Details
Sep 24, 2025, 10:51 PM	#11	Allocated - Rajajinagar Sports Complex Shelter	Demo User	Details
Sep 24, 2025, 12:18 PM	#10	Allocated - Rajajinagar Sports Complex Shelter	Demo User	Details
Sep 24, 2025, 12:17 PM	#9	Allocated - Rajajinagar Sports Complex Shelter	Demo User	Details
Sep 24, 2025, 10:07 AM	#8	Allocated - Jayanagar Civic Assembly Hall	Demo User	Details

IX. LIMITATIONS

- 1) All data is fictional and for Bengaluru only. There is no link to a real shelter registry.
- 2) Ranking uses straight-line distance. Road distance is shown only after a route is requested.
- 3) A recommendation does not reserve space, so two people can be sent to the same last spaces.
- 4) Occupancy changes only when an administrator updates it.
- 5) Routing depends on the public OSRM demo server, which has no uptime guarantee.
- 6) The admin screen for verifying and activating potential facilities is not built yet, though the backend works.
- 7) The score weights and limits are fixed values we chose. We have not tested them on real disaster data.
- 8) Exclusion reasons are returned by the API but not yet shown in the interface.
- 9) Google sign-in and phone sign-in do not work yet. Email and password sign-in works.
- 10) The API and browser test scripts have not been updated for the latest changes.

X. FUTURE SCOPE

- Reservation and check-in so occupancy stays accurate.
- Ranking by road travel time instead of straight-line distance.
- Self-hosted routing.
- Splitting a group across shelters when none can take everyone.
- Live occupancy updates, offline support, and Kannada, Hindi and English screens.
- SMS access for people without a smartphone.
- A shelter-operator role limited to one shelter.
- The admin screen for the facility verification and activation flow.

XI. CONCLUSION

We built a web application that recommends emergency shelters. It first removes the shelters that cannot be used and then ranks the rest with a score we can explain. In our tests on the demo data, a farther shelter with more space was ranked above a nearer shelter that was almost full, and the ranking changed as expected when we changed the priority, disaster type and group size. The current version uses only demo data. The main work left is the admin screen for verifying facilities, reserving spaces, and testing with real shelter data.

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